

Question Number	Answer	Additional Guidance	Mark
2a	An answer that includes the following: • A divide to form the male (gamete / nuclei) (1) • B produces (digestive) enzymes / digests the style / makes pathway (for male gamete / to ovum) /grows towards ovum (1)	ACCEPT ovary, ovule, female gamete, egg	expert 2

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2b	An answer that includes the following points: • credit suitable methods used (to observe germination of pollen grains) (1) • sucrose solution added (1) • suitable time quoted before germination rate assessed (1) • control of one relevant abiotic variable (1) • control of one relevant biotic variable (1) • view through microscope / count germinated & non-germinated / count germinated & total (and work out percent) (1) • reference to relevant health and safety issue (1)	e.g. hanging drop, agar plates, filter paper, cavity slide ACCEPT boron / sugar. > or = 15 minutes < or = 5 days temperature, pH, light, sucrose concentration, fixed time species, a cultivar, variety of plant, type species of bee e.g. pollen allergy, bee stings	Expert 5

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2ci	An answer that includes the following points: - the optimum is at higher temperature for bee than hand collected pollen / described / both optima quoted (30C and 36C) (1) - the percentage germination is higher in hand (than bee collected pollen at all temperatures) (1) - range in percentage germination in hand greater than in bee collected / fall from optimum {higher / steeper} in hand than bee collected / temperature has greater effect on hand collected (1) - enzymes denature (above the optimum / above 30C and 36C in both) (1) - there are no {error bars / SDs / SEs} (1)	ACCEPT both temperatures at which % germinations starts to go down quoted ACCEPT reverse arguments ACCEPT ranges quoted (-3 or -4 and -16) from 22 to 19/18 56 to 40/41 etc. or peak to lowest ACCEPT there is no measure of variability	Expert 4

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2cii	An answer that includes the following points: • repeat the experiment at 30C (1) • {all conditions / named condition} should be kept constant (1) • (mean and) SD (calculated) / error bars (1) • look for overlap in SDs / perform t-test (1)	IGNORE the number of repeats IGNORE temperature ACCEPT named relevant other test of difference	Expert 4

Question Number	Answer	Additional Guidance	Mark
2di	A calculation with the following steps: - lengths correctly read from graph (1) - subtraction followed by correct division using data above and multiplication by 100(1) - quoted to correct sig figs. and as a percentage (1)	Correct answer gets 3 marks with no working Ecf at mp 2 and 3 1000 and 750 1000 - 750 = 250 then $250 \div 750 = 0.3333$ OR 1000 - 750 = 250 $250 \div 1000 = 0.25$ OR $250 \div 875 = 0.2857$ 3.3 (%) 25.0 (%) 28.6 (%) Accept 25, 33 or 29 for 2 marks	Expert 3

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2dii	pollen damaged (in bee sample)/ description (1)	ACCEPT reverse argument. Allow secrete something which affects pollen negatively/ collect less ripe pollen / bees selective / bee pollen from a different species	expert 1